Human Questions

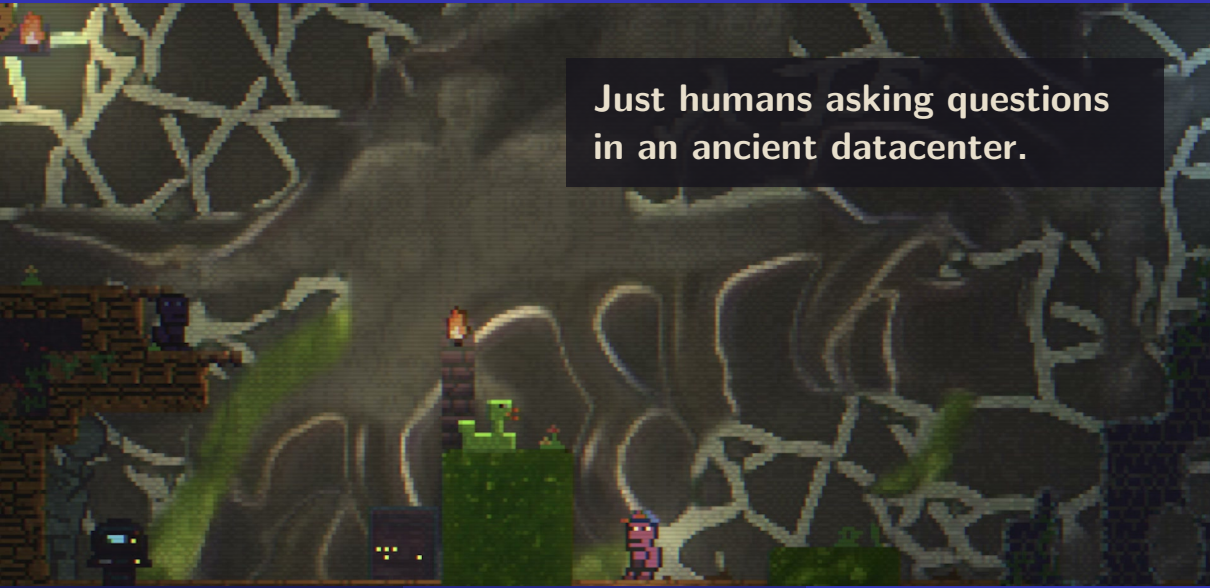
Toward an Appetizer-Sized, Knowledge-Gated Exploration Game

K. T. Watters[1,2,3,4,5,6]

[1]Picea Abies Institute for Engine Technology, Indiana [2]Conservatorio Nocturno de Música, Buenos Aires
[3]Berliner Institut für Pixelkunst e. V. [4]Kyoto Center for Garden and Level Design
[5]Arbeitsgruppe Neuronale Grafik, Berlin [6]Office of Finance (vacant)

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Graphical Abstract



Abstract

We present *Human Questions*, an ancient datacenter dive with a partner in crime: an appetizer-sized, knowledge-gated exploration game with juicy combat and unconventional traversal mechanics.

We propose to finish and release the single-player game on Steam by Q2 2027 with a single author ($n = 1$) and a budget of $50,000.

Preliminary results from an in-person showcase[1] suggest that gliding along walls as a stream of photons is fun ($p < 0.05$). One test subject had to be dragged away from the gamepad.

Keywords: metroidbrainia; burrowing; neural cellular automata; ducks

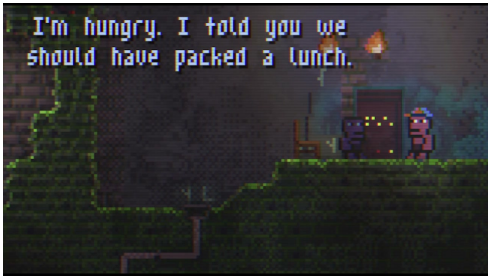


Figure 1: A partner in crime (left) and the protagonist (right).

[1] Showcased at Talk & Play #57 in Berlin.

Results

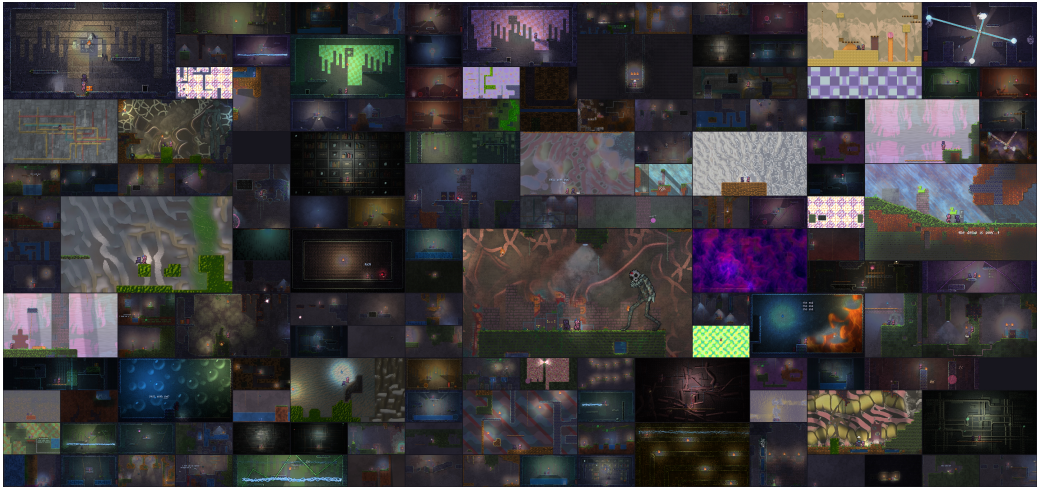


Figure S1: Rooms sampled uniformly at random. (They were not.)

Inclusion and Exclusion Criteria

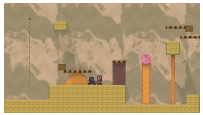
What Stays

- Dense, intricately connected world demanding attention [1, 2]
- Mood, mystery, atmosphere, a certain lightness [3]
- Music at the emotional core (composed in-house, $n = 1$)
- Respect for: player's time and intelligence

Non-goals

- Hand-drawn vector-style visuals [2]
- Lock-and-key ability gating [4]. The key to the "door" is you, noticing.
- Busy parallax; instead favoring flat, painterly compositions: lo-fi 2D, glammed up for 2027





Criteria were established prior to data collection, which consisted of playing a very large number of games.

[1] Nigoro, *La-Mulana* (2012) [2] Team Cherry, *Hollow Knight* (2017) [3] Mobius Digital, *Outer Wilds* (2019) [4] Shared Memory, *Animal Well* (2024)

Methods

North star: the character controller. Hit-stun and i-frames millisecond-tuned; expressive moves chain into each other; approachable, with a high skill ceiling. Everything flows from that!

Burrow Dive into any surface and swim through it; pop out with momentum.

Ride Burrow lines are express tunnels. Some invert the room's geometry.

Move Ropes, conveyors, contraptions, bouncers.

Fight Readable combat and bosses which test your grasp of the expressive moveset.

Notice Knowledge gates: the way forward is on screen before you understand it.

Apparatus: a custom engine and in-game editor for that "and now for something (mostly) different" vibe

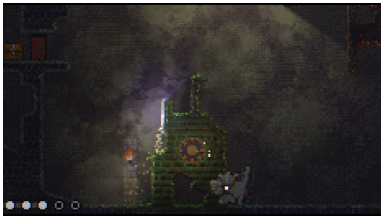


Figure 2a: Burrowing.

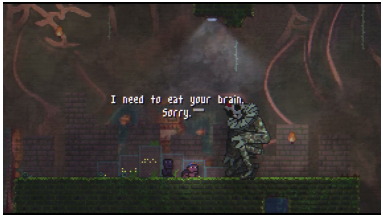


Figure 2b: Combat, when necessary.

Materials

The datacenter is alive. Its surfaces are painted by hand and grown by neural cellular automata: grids of tiny cells, each running the same small learned rule.

Nobody animates them. Hit one, and it has to heal. The cells are part of the gameplay: swim through them, find a way to use them against your foes...

Instruments: live textures, dynamic lighting, shaders everywhere.

Trained in-house on rented GPUs, building on Cells[2]Pixels[2] (used with the authors' permission).

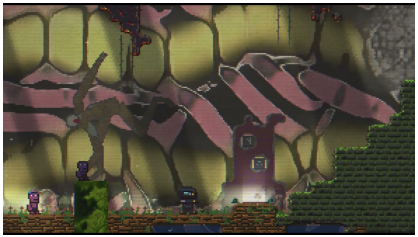


Figure 3: A self-organizing material. No two frames alike.



Figure 4: The study site, as currently surveyed.

[2]Pajouheshgar et al., *From Cells to Pixels*, SIGGRAPH 2026. `https://cells2pixels.github.io/`

Author Contributions

Role	Contributor
Conceptualization	K.T.W.
Engine, editor & tools	K.T.W.
Music	K.T.W.
Pixel art, tiles & backgrounds	K.T.W.
Level design	K.T.W.
Neural textures	K.T.W., Cells^{2}Pixels
Sound design	K.T.W., freesound.org
Funding acquisition	*this slide*

Since May 2026: 1,033 commits, 99 rooms, one level editor, a public showcase.

Experience: programming since 199X; games played, $n \gg 1$.

Selected prior work

[1] *Zero Days VR*. With Scatter. Emmy winner.

[2] *GodotVision*. Godot on Apple Vision Pro.

[3] *Threezy*. 3D art on your phone.

[4] *Content Creator Simulator*; *A Very Holographic Aquarium*. Looking Glass.

[5] *Raymarching Toolkit for Unity*.

[6] *Strange Dates*. Queerness and Games Conference, 2018.

[7] *Wish U Were Here*. With R. Yang, E. Cameron, F. Ramallo.

More at `https://kev.town`

Response to Reviewers

Reviewer 2: "The player character art reads as unfinished vs the backgrounds."
Response: Agreed. A character art pass is scheduled.

Reviewer 3: "Which pits are navigable and which are death is unclear."
Response: We are removing the pits. You can go everywhere.

Reviewer 4: "He thought the duck would hurt you."
Response: We stand by the duck.

Limitations. Scope is one world, single-digit-hour-count sized.

Budget & Declarations

Table 1: Proposed budget.

Salary & living expenses (9 months)	$50,000
Total	**$50,000**

Sep 2026	Public prototype (done)
Q4 2026	First area complete; Steam page
Q1 2027	Content complete; demo
Q2 2027	Launch on Steam

Funding. This work is not currently supported by any grant. (Yet.)

Conflicts of interest. The author would very much like to make this game.

Data availability.
`https://keelo.itch.io/humanquestions`

Correspondence. kevinwatters@gmail.com

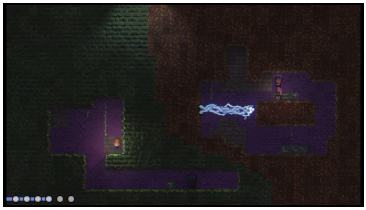


Figure 5: Robustness check: results hold inside out.